Reda Karam

AI & Bioinformatics

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Summary

AI and Biomedical Engineering student passionate about Bioinformatics, Computational Biology, and Healthcare AI. Proficient in biological data analysis, genomics, and software development, with hands-on experience in building AI models for medical applications. Eager to leverage computational skills in an AI & Bioinformatics Internship to solve complex biological challenges

Education

Bachelor's in Artificial Intelligence at Horus University, New Damietta, Egypt **August 2024 - July 2028**

Internships

Ai &Data Science Trainee at Digital Egypt Pioneers Initiative (DEPI), Egypt **July 2026 - Present**

completed intensive practical training focused on Data Science and Machine Learning fundamentals. Applied Python, Pandas, NumPy, and Matplotlib for data preprocessing, visualization, and exploratory data analysis (EDA). Built and evaluated predictive Machine Learning models using Scikit-Learn.

Artificial Intelligence Trainee at National Telecommunication Institute (NTI), Egypt **May 2026 - June 2026**

Completed specialized training in Artificial Intelligence algorithms and practical implementations. Developed AI models using Python, focusing on feature selection, model tuning, and accuracy assessment. Participated in hands-on projects implementing core Machine Learning techniques

Bioinformatics Intern at Biotech World Company, Egypt **May 2026 - Present**

- Performed hands-on biological data analysis, sequence alignment (BLAST), and genomic data processing using modern bioinformatics tools.
- Developed custom Python scripts to automate biological data extraction, preprocessing, and pipeline analysis.
- Applied computational biology methods and molecular visualization tools to analyze protein-ligand interactions and nucleic acid structures.
- Collaborated with the computational team on genomic variant analysis and interpreting biomedical datasets for practical applications.

Skills

Programming Languages: Python, C++, SQL, R, HTML5 & CSS3

Bioinformatics & Computational Biology: Bioinformatics Data Pipelines, Biopython, Genomic Data Analysis, Sequence Alignment (BLAST), Biological Data Mining, Protein & Nucleic Acid Analysis

Data Science & Analytics: Exploratory Data Analysis (EDA), Data Cleaning & Preprocessing, Data Visualization, Pandas, NumPy, Matplotlib & Seaborn, Statistical Analysis

AI & Machine Learning: Artificial Intelligence, Machine Learning Algorithms, Model Evaluation & Tuning, Feature Engineering, Predictive Modeling, Scikit-Learn

Tools & Frameworks: Git, GitHub, Streamlit, Linux / Bash Scripting

Soft Skills: Problem Solving & Logic, Analytical Thinking, Teamwork & Collaboration, Technical Communication, Research & Continuous Learning

Projects

Lung Cancer Risk Prediction Model

- Developed a predictive Machine Learning model to assess human lung cancer risk based on clinical and lifestyle questionnaires.
- Performed exploratory data analysis (EDA), data cleaning, and feature selection to optimize model performance and accuracy.
- Built an interactive interface for users to input responses and receive instant, data-driven health risk evaluations

Genomic Sequence & Mutation Analyzer

- Built a Python-based bioinformatics pipeline for processing DNA/RNA sequences and analyzing genomic mutations.
- Implemented sequence alignment algorithms (BLAST) and integrated Biopython to extract genetic variants and sequence motifs.
- Created interactive visualization tools to display mutation distributions and biological sequence properties efficiently.

Task Manager Application

- Developed a functional Task Manager application designed to track, organize, and prioritize daily tasks and projects.
- Implemented CRUD operations (Create, Read, Update, Delete) with state management for smooth task workflow.
- Designed a clean and responsive user interface to enhance productivity and task tracking efficiency.

Languages

English - Conversational

Certificates

Fundamentals of WGS Data Analysis from Genotech — **November 2025**
Practical Molecular Dynamics Workshop from Genotech — **March 2025**
Industrial Mycology from BGE Journey — **July 2025**
Certificate of Membership from InformaTalents — **July 2025**
Python: Programming Fundamentals I& II from Almdrasa — **January 2026**
AI For You from Oracle — **February 2026**
Overview of AI from Huawei ICT Academy — **May 2026**
Introduction to AI from Almdrasa — **February 2026**